AN EULER CHARACTERISTIC FORMULA FOR THE COTANGENT COMPLEX OF A SQUARE OF FIELDS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Jacobi-Zariski exact sequence associated with a tower of field extensions. used twice at proof lines 8645-8665. Supplies the two exact sequences whose alternating sums are compared.
- Finiteness of the cotangent spaces and of the second homology for finitely generated extensions and a formula for their alternating sum. the author cites Lemma lemma-cartier-equality at proof lines 8666-8670. Provides the numerical input that turns the comparison into the stated formula.
- Identification of the kernels and cokernels of the comparison maps inside the Jacobi-Zariski sequences. used at proof lines 8650-8666. Locates the terms whose dimensions appear in the statement.

Source reference. Original lemma 07E3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9433–9433. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07E1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9431–9431. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Given a commutative diagram of fields*

$$\begin{array}{ccc} K & \longrightarrow & K' \\ \uparrow & & \uparrow \\ k & \longrightarrow & k' \end{array}$$

with k'/k and K'/K finitely generated field extensions the kernel and cokernel of the maps

$$\alpha : \Omega_{K/k} \otimes_K K' \to \Omega_{K'/k'} \quad \textit{and} \quad \beta : H_1(L_{K/k}) \otimes_K K' \to H_1(L_{K'/k'})$$

are finite dimensional and

$$\dim \mathrm{Ker}(\alpha) - \dim \mathrm{Coker}(\alpha) - \dim \mathrm{Ker}(\beta) + \dim \mathrm{Coker}(\beta) = \mathit{trdeg}_k(k') - \mathit{trdeg}_K(K')$$

Proof. The Jacobi-Zariski sequences for $k \subset k' \subset K'$ and $k \subset K \subset K'$ are

$$0 \to H_1(L_{k'/k}) \otimes K' \to H_1(L_{K'/k}) \to H_1(L_{K'/k'}) \to \Omega_{k'/k} \otimes K' \to \Omega_{K'/k} \to \Omega_{K'/k'} \to 0$$

and

$$0 \to H_1(L_{K/k}) \otimes K' \to H_1(L_{K'/k}) \to H_1(L_{K'/K}) \to \Omega_{K/k} \otimes K' \to \Omega_{K'/k} \to \Omega_{K'/K} \to 0$$

By Lemma 07E1 the vector spaces $\Omega_{k'/k}$, $\Omega_{K'/K}$, $H_1(L_{K'/K})$, and $H_1(L_{k'/k})$ are finite dimensional and the alternating sum of their dimensions is $\mathrm{trdeg}_k(k') - \mathrm{trdeg}_K(K')$. The lemma follows. □